Vector Calculus and Complex Analysis Classwork (2025/2/17)

Name:

Student ID:

Let $\boldsymbol{v} = (\sqrt[3]{xy}, 2^{yz}, \cos(xz))$. Calculate grad($\boldsymbol{v}$), div($\boldsymbol{v}$) and curl($\boldsymbol{v}$) if possible. Explain if it is not possible to calculate.

Solution

$$\begin{aligned}
\operatorname{div}(\boldsymbol{v}) &= \frac{\partial}{\partial x}\sqrt[3]{xy} + \frac{\partial}{\partial y}2^{yz} + \frac{\partial}{\partial z}\cos(xz) \\
&= \frac{1}{3}\sqrt[3]{x^{-2}}\sqrt[3]{y} + z2^{yz}\ln 2 - x\sin(xz) \\
\operatorname{curl}(\boldsymbol{v}) &= \begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \sqrt[3]{xy} & 2^{yz} & \cos(xz) \end{vmatrix} \\
&= \left(\frac{\partial}{\partial y}\cos(xz) - \frac{\partial}{\partial z}2^{yz}\right)\boldsymbol{i} + \left(\frac{\partial}{\partial z}\sqrt[3]{xy} - \frac{\partial}{\partial x}\cos(xz)\right)\boldsymbol{j} + \left(\frac{\partial}{\partial x}2^{yz} - \frac{\partial}{\partial y}\sqrt[3]{xy}\right)\boldsymbol{k} \\
&= \left(-y2^{yz}\ln 2\right)\boldsymbol{i} + \left(z\sin(xz)\right)\boldsymbol{j} + \left(\frac{1}{3}\sqrt[3]{y^{-2}}\sqrt[3]{x}\right)\boldsymbol{k}
\end{aligned}$$

It is **NOT** possible to calculate grad($\boldsymbol{v}$) because $\boldsymbol{v}$ is a vector function, but not a scalar function.